










# Information Visualization

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# Introduction

# *Introduction*

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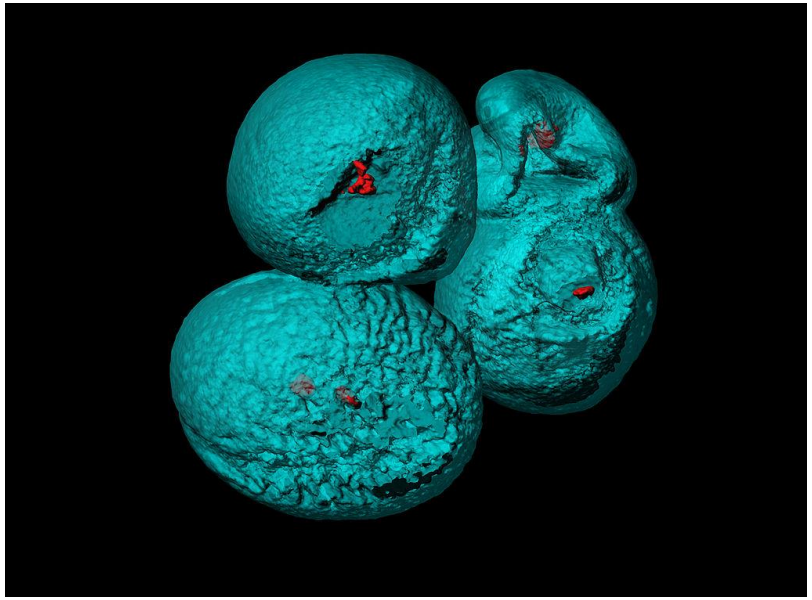
- The human ability to detect patterns and trends.

- **Scientific visualization** - data associated with a spatial position, typically physically based.
- **Information visualization** - data without an assigned spatial position, non-physically based.

- Two main goals:
  - communicate or record a certain event or process
  - analyze the data

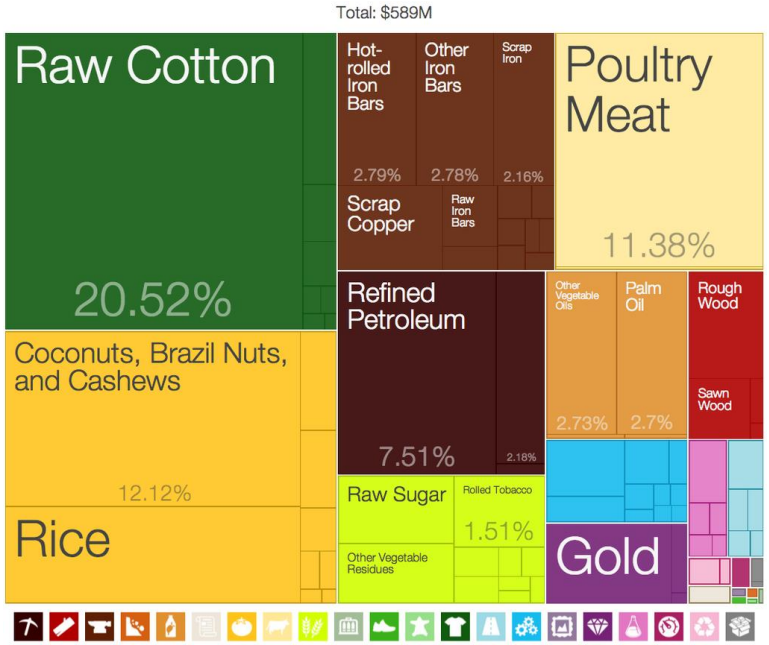
# *Introduction*

## Scientific visualization



Wikimedia Commons. (2020). File:Tolmuterad.jpg — Wikimedia Commons, the free media repository.

## Information visualization




Wikimedia Commons. (2020). File:Benin English.png — Wikimedia Commons, the free media repository.

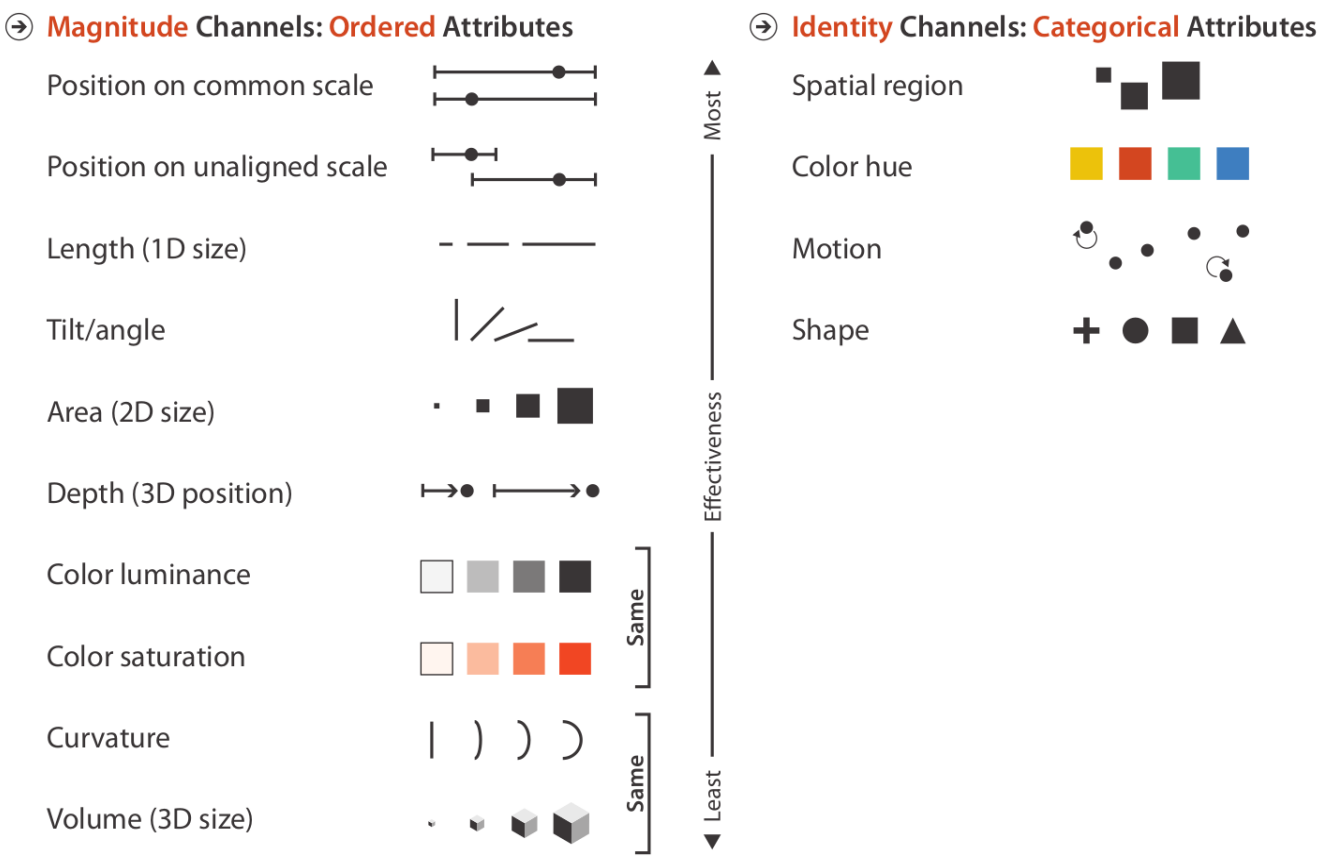
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There are 4 main types of data:

| Type | Operations | Example |
|---|---|---|
| Nominal | ==, != | Smoker/non-smoker |
| Ordered | ==, !=, <, >, <=, >= | Minimal/moderate/severe pain |
| Discrete | ==, !=, <, >, <=, >= and arithmetic operations | Number of visits, number of cells |
| Continuous | ==, !=, <, >, <=, >= and arithmetic operations | Height in cm, temperature in K |

# Introduction




Munzner, T. (2014). *Visualization Analysis and Design (AK Peters Visualization Series)*. A K Peters/CRC Press.

# Quality metrics

# *Quality metrics*

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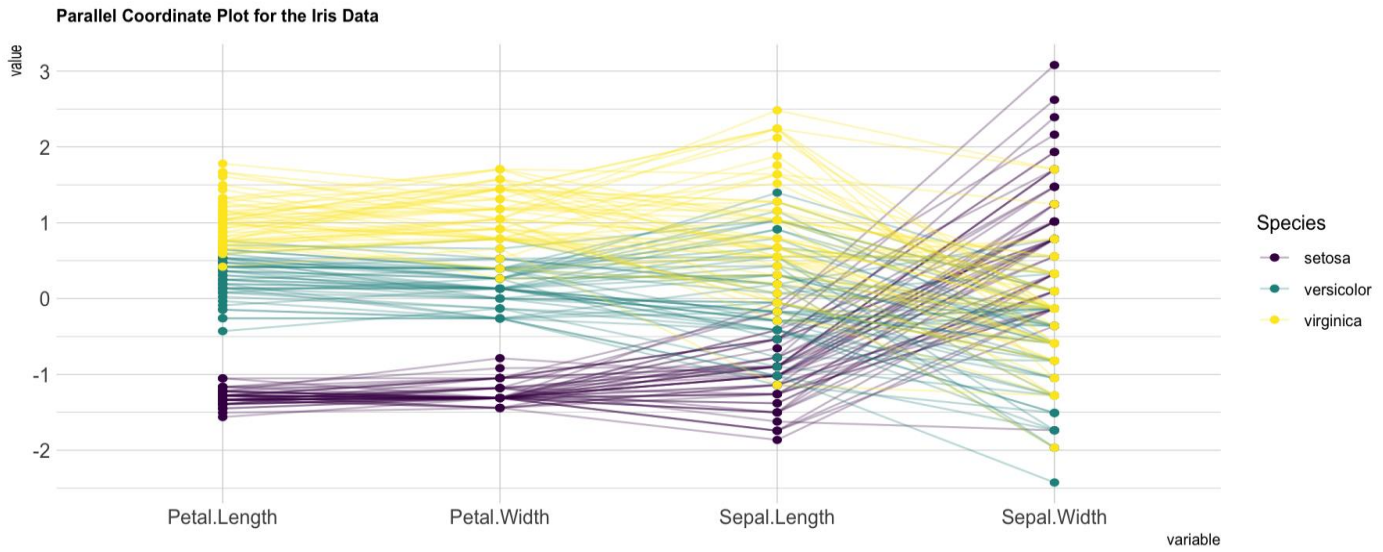
- The most common way is a user study.
- There are attempts to use information theory to evaluate visualization performance.
- Four types of information content measures:
  - the amount of information spanned by **the data**
  - the amount of information spanned by **the display**
  - the amount of information in **a particular data display**
  - the amount of **topological information** content

Yang-Peláez, J., & Flowers, W. (2000). Information Content Measures of Visual Displays. In *Proceedings of the IEEE Symposium on Information Vizualization 2000* (pp. 99). IEEE Computer Society.

# Parallel coordinates

# *Parallel coordinates*

- Parallel axes for different data variables
- **Pro**: easily detectable patterns between **consecutive** axes
- **Con**: not so easily detectable patterns between **separated** axes

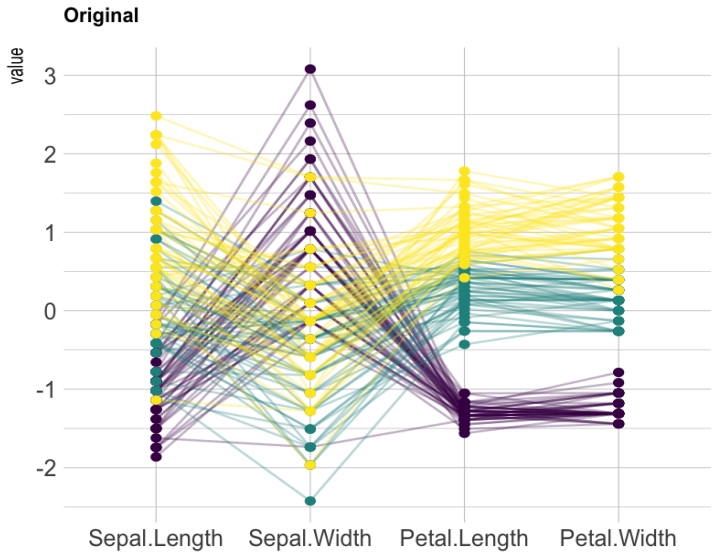



Healy, Y. H. and C. (n.d.). *Parallel coordinates plot*. Parallel coordinates plot – from Data to Viz. https://www.data-to-viz.com/graph/parallel.html#.

# *Parallel coordinates*

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Axis ordering is very important.



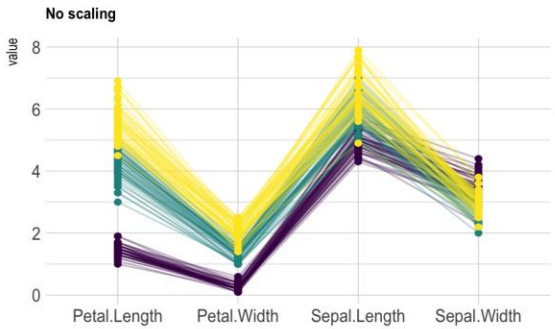


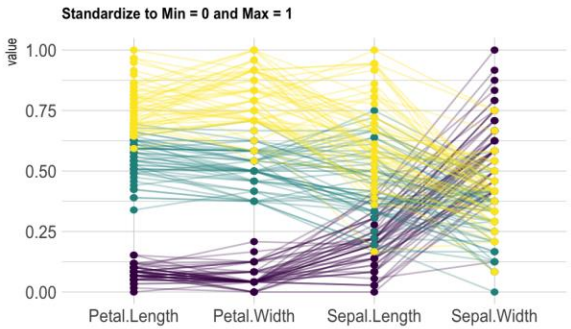


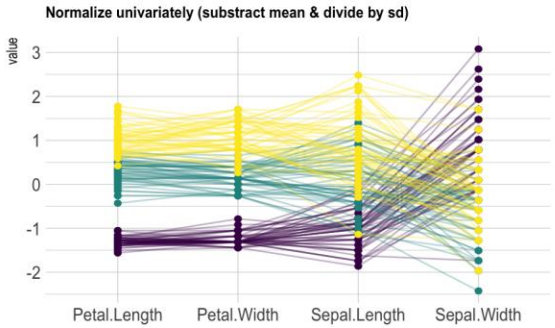
Healy, Y. H. and C. (n.d.). *Parallel coordinates plot*. Parallel coordinates plot – from Data to Viz. https://www.data-to-viz.com/graph/parallel.html#.

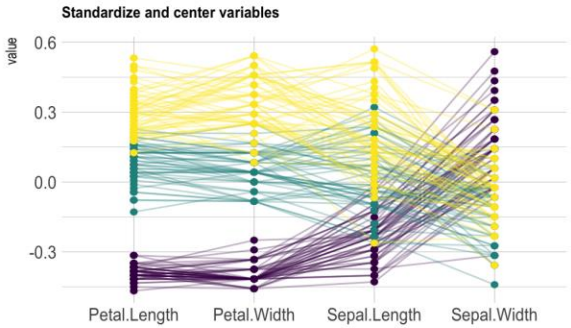
# *Parallel coordinates*

Scaling is also important.




Healy, Y. H. and C. (n.d.). *Parallel coordinates plot*. Parallel coordinates plot – from Data to Viz. https://www.data-to-viz.com/graph/parallel.html#.

# *Parallel coordinates*

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- How to solve axis ordering?
  - Find highly related variables and set them as neighbours.
- How to find those variables?
  - The Pearson correlation coefficient (assumes a linear relationship)
  - The mutual information equation
- How to evaluate our visualization?
  - Calculate the entropy of a rendered image and look only at gray levels.

High entropy = many gray pixels = many line crossings

# Multivariate data

# *Multivariate data*

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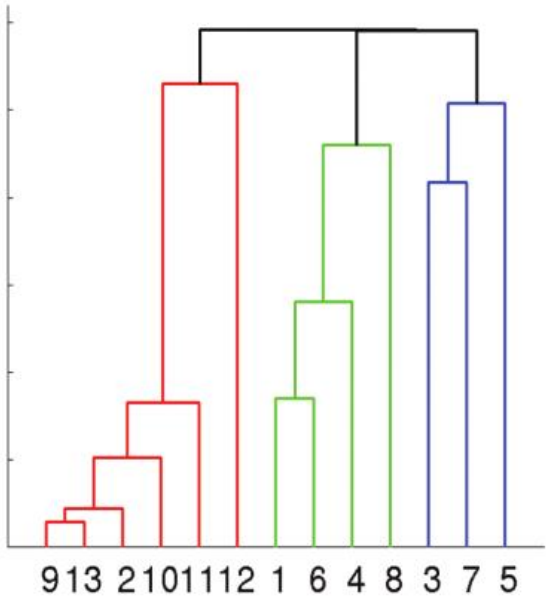
- Most real-world phenomena are dependant on multiple factors
- To understand a dataset, we have to inspect the relationships among the variables.
- This is **one of the hardest tasks** in information visualization.

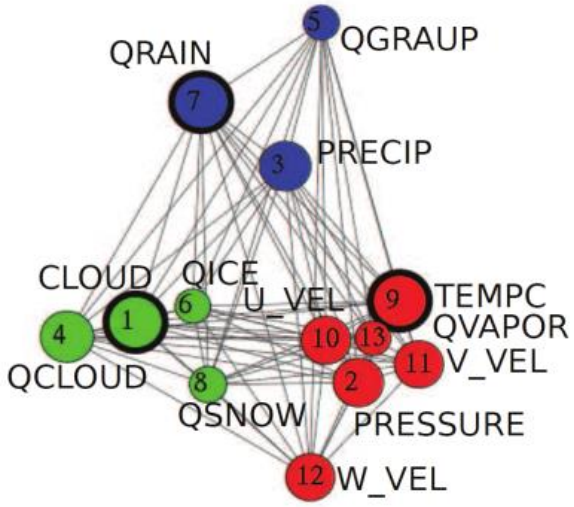
- There are 2 ways to get some knowledge about variables:

  - Use Shannon entropy, and specifically joint entropy, to compute entropy for different clusters of variables
  - Calculate mutual information between different variables (it can measure non-linear relationships as well as linear)

# *Multivariate data*

Two different ways to visualize mutual information: dendrogram, graph.




A. Biswas, S. Dutta, H. Shen, & J. Woodring (2013). An Information-Aware Framework for Exploring Multivariate Data Sets. *IEEE Transactions on Visualization and Computer Graphics, 19(12), 2683-2692.*

# *Multivariate data*

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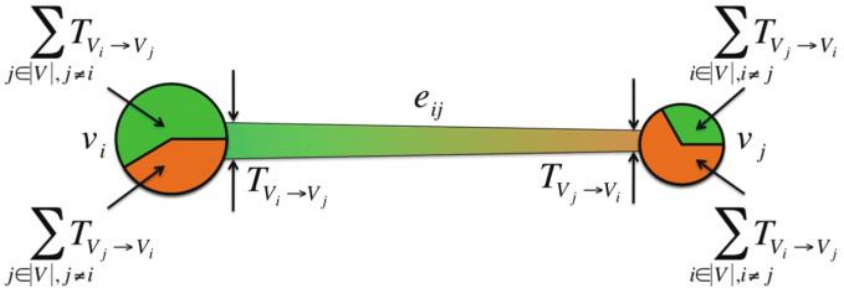
- The even harder problem appears if we have **time-varying multivariate data**.
- We have to inspect relationships between multiple variables considering now the time dimension.

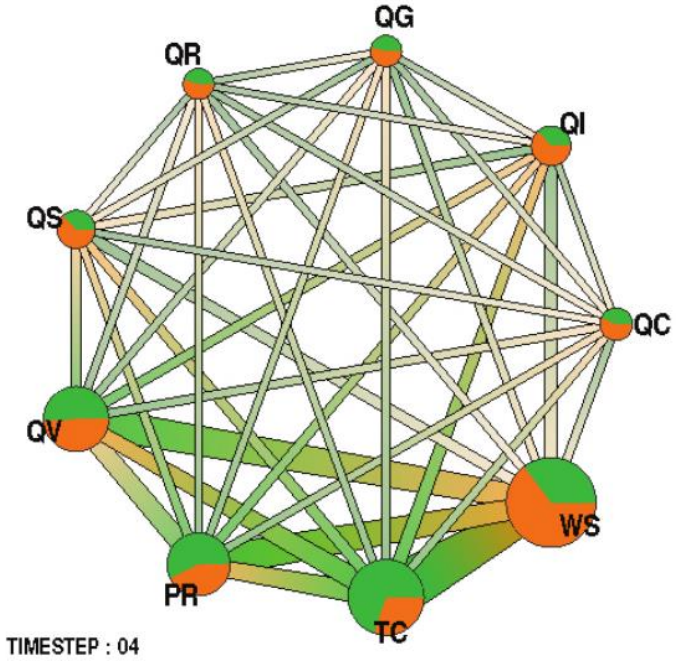
- Two popular techniques to estimate relationships:
  - Cross-correlation
  - Mutual information

- These techniques are symmetric and not appropriate for assessing causality.
- Better but more complex technique - **transfer entropy**.

# Visualization of Information Transfer

# Visualization of Information Transfer




Chaoli Wang, Hongfeng Yu, Grout, R., Kwan-Liu Ma, & Chen, J.. Analyzing information transfer in time-varying multivariate data.

# Privacy and uncertainty

# *Privacy and uncertainty*

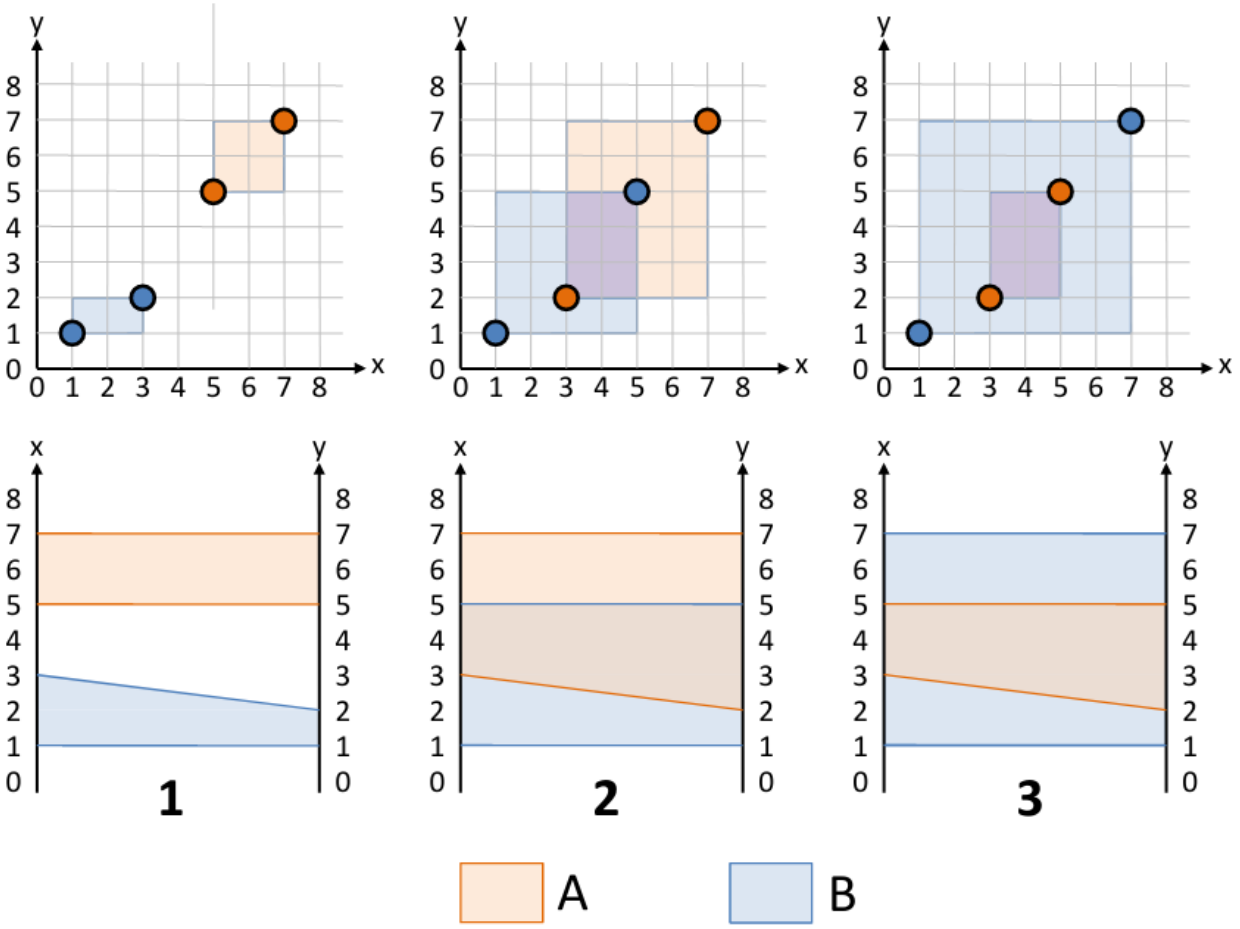
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- Sometimes we want to hide original data and visualize only a representation (encoding) of that data.
- Healthcare records, bank records...
- If we don't do this, someone could identify supposedly anonymized individuals.

Increasing visual uncertainty, increases privacy of visualization but decreases its utility.

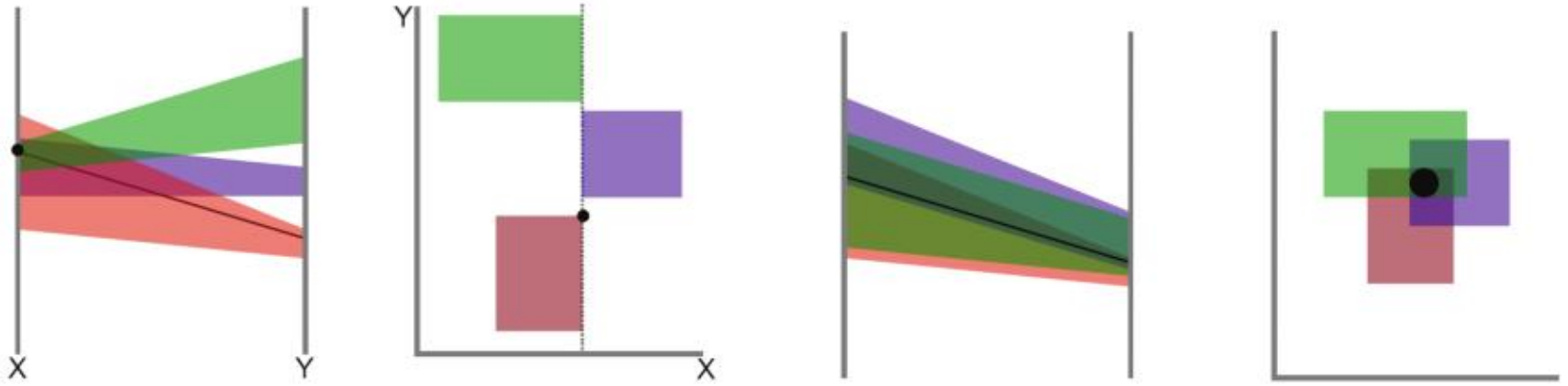
- There are multiple metrics of privacy in cluster-based parallel coordinates plots.

# Privacy and uncertainty




Dasgupta, A., Chen, M., & Kosara, R. (2013). Measuring Privacy and Utility in Privacy-Preserving Visualization. *Computer Graphics Forum, 32(8),* 35-47.

# *Privacy and uncertainty*




Dasgupta, A., Chen, M., & Kosara, R. (2013). Measuring Privacy and Utility in Privacy-Preserving Visualization. *Computer Graphics Forum, 32(8),* 35-47.

Thank you for your attention!